Reference, task policy and actuator realization



roadside operating context

Reference generator
verge line + cut height

Vehicle layer
constant road speed

DLS teacher
privileged; training only

distillation / identification

TASK POLICY

terrain profile · task error · state · m

66-D input → **MLP** hidden → **MLP** hidden → **4-D action** desired motion

LEARNED ACTUATOR REALIZATION — one lane active per experiment

PPO–ROM

Command-conditioned ROM
GRU + hydraulic update

PPO–ID–FD

ID
motion → valve

FD
valve → state

PPO–ID(AAM)–FD

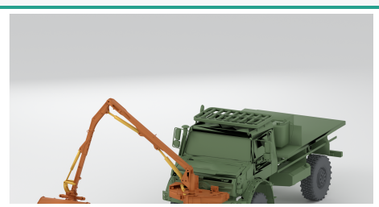
ID
frozen

AAM
zhat

adapter
bounded

FD
frozen

Bounded PhysX target
same for all stacks



USD / PhysX plant

measured cylinder state and actuator-model error